

EXP. NO: 10

PERFORMANCE ANALYSIS OF FM SYSTEM UNDER GAUSSIAN

Objectives

1. To generate a Frequency Modulated (FM) signal in MATLAB using the specified carrier and message signal parameters.
2. To introduce Additive White Gaussian Noise (AWGN) into the generated FM signal and simulate the effect of a noisy communication channel using MATLAB.
3. To analyze the effect of AWGN on the transmitted FM signal by comparing the original and noisy FM signals using time-domain and frequency-domain analysis.

CODE

```
clc;
clear;
close all;

% Parameters
fs = 100000;    % Sampling frequency
t = 0:1/fs:0.02; % Time vector
fm = 500;       % Message frequency
fc = 5000;      % Carrier frequency
kf = 100;       % Frequency sensitivity (Hz/V)
SNR = 10;       % SNR in dB

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% FM Modulation
fm_sig = cos(2*pi*fc*t + ...
    2*pi*kf*cumsum(m)/fs);

% Add Gaussian White Noise
signalPower = mean(fm_sig.^2);
```

```

noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower)*randn(size(fm_sig));
noisy = fm_sig + noise;

% Display Results
figure('Name','FM Modulation with Noise', ...
    'NumberTitle','off');

subplot(4,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,3);
plot(t,fm_sig);
title('FM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,4);
plot(t,noisy);
title('FM Signal with Gaussian White Noise (10 dB SNR)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Display information
fprintf('Message frequency = %d Hz\n',fm);
fprintf('Carrier frequency = %d Hz\n',fc);
fprintf('Frequency sensitivity = %d Hz/V\n',kf);
fprintf('SNR = %d dB\n',SNR);

```

OUTPUT

